

Education

Vellore Institute of Technology	Chennai, India
B.Tech in Computer Science and Engineering; CGPA: 8.54	Aug 2023 – Jul 2027
Government Higher Secondary School	Class XII: 92.5% 2021–2023
Devi Matriculation Higher Secondary School	Class X: Pass 2020–2021

Skills Summary

Languages: Python, C++, Java, JavaScript, Solidity, SQL
Frameworks: React.js, Node.js, Express.js, Next.js, Tailwind CSS, Streamlit
Databases & ORM: MongoDB, PostgreSQL, OracleSQL, Prisma
Libraries: Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, PyTorch
Tools: Git, GitHub, Docker, Postman, AWS
Technologies: Machine Learning, Generative AI, Smart Contracts, Blockchain (Solidity), Cloud Deployment

Projects

- ChainFund – Decentralised Risk-Aware Community Lending Protocol**
Solidity, Foundry, React.js, ethers.js, Node.js, Gemini API
- Built a smart-contract chit fund where members deposit periodically and bid via sealed-bid auctions to receive the pool, eliminating organiser fraud common in traditional informal lending.
 - Implemented on-chain repayment tracking and an AI-powered credit oracle using Gemini API with a pure-math fallback to score members based on repayment history.
 - Deployed to Ethereum Sepolia testnet using Foundry; integrated React + ethers.js v6 frontend with a Node.js oracle service.
- Multimodal Deepfake Detection System**
Python, PyTorch, OpenCV, SyncNet, Scikit-learn
- Designed a two-branch deepfake detector combining lip-sync inconsistency analysis and frame-level temporal artifact detection to identify AI-generated fake videos.
 - Integrated a SyncNet-style audio-visual matching model alongside a CNN-based frame encoder and fused outputs using an MLP classifier for final prediction.
 - Added an explainability layer returning human-readable reasoning such as audio-lip mismatch scores and frame artifact indicators.
- Neuro-Symbolic Tax Optimisation Engine**
Python, Streamlit, Scikit-learn, Pandas, NumPy
- Built a hybrid reasoning system combining symbolic rule-based tax logic with ML-driven heuristics to generate personalised tax optimisation recommendations.
 - Parsed structured financial inputs against tax regulation rules and applied trained models to identify deduction opportunities and liability reduction strategies.
 - Deployed as an interactive Streamlit web app enabling users to simulate tax scenarios and visualise savings in real time.

Work Experience

Student Council Member	2025 – Present
Vellore Institute of Technology (VIT), Chennai	
- Managing and coordinating student affairs across departments - Proposing improvements to campus discipline and student welfare policies - Facilitating communication between the student body and college administration	
Active Member, Artificial Intelligence Club	2024 – Present
Vellore Institute of Technology (VIT), Chennai	
- Contributed to collaborative AI-based projects and research discussions - Assisted in organising AI workshops, seminars, and technical events	

Certifications

Introduction to Generative AI – IBM	2024
Fundamentals of DevOps – IBM	2024
Generative AI – Oracle	2025